Originality of the results

1. Statements not found in the literature as written

The method behind each is published; these are original *statements* obtained by specialising it, not a new theory of branching. Each was searched against the chapter bibliographies and the wider literature, then challenged adversarially.

Where	The statement	Nearest published work
Thm 6.8 p. 167	**The flooding identity.** At matched lifetime yield the extinction probabilities of bursting and budding differ by exactly $\Delta z_{\text{ext}} = (L-1)(1/m-1)$, so bursting favours establishment precisely when $L > 1$, equivalently when $1/\delta < 1/\lambda + 1/\mu$.	Pearson, Krapivsky and Perelson (2011) find $p_v^{\text{burst}} \le p_v^{\text{cont}}$ — bursting always favoured — from a fixed burst size, with no parameter that can reverse the order.
Prop 6.9 p. 169	**Coincidence of the offspring laws.** At $L = 1$ the bursting and budding offspring *distributions* coincide, not merely their extinction probabilities: the burst generating function collapses to the geometric $(1-\varrho)/(1-\varrho z)$. The variance gap $2q^2V_\infty(L-1)/(a-1)$ changes sign with $L-1$.	Pearson's geometric comparator; Karlin and Tavaré's geometric killing position. Neither yields a coincidence of laws.
Thm 7.4 p. 193	**Exact finite-time survival under two-type killing.** A closed form for the no-catastrophe probability of the triangular two-type birth–death process with type-specific catastrophe rate $\delta_1 X + \delta_2 Y$, as a Wronskian mixture of a real-axis ${}_2F_1$ pair.	Antal and Krapivsky (2011) apply the same Gauss reduction to terminal counts without killing. Setting $\delta = 0$ here forces $S \equiv G \equiv 1$ and does *not* return their generating functions.
Eq. 7.50 p. 205	**The killed bivariate generating function.** The defective PGF under the same killing, by the same Wronskian mixture. This is the object for which $\delta_i = 0$ *does* recover Antal–Krapivsky, and it is what makes the previous entry a different observable rather than a reparametrisation.	Antal and Krapivsky (2011), eqs. (25)–(29), recovered here as the $\delta_i \to 0$ limit.

2. Extensions: an ancestor, and what is added to it

Ordered by chapter. In each case a published result is the ancestor; the thesis changes the process, the object, the hypotheses, or the biological reading.

Where	The increment	Ancestor
§ 2.2.4.1 p. 38	Gated release in which the opening intensity ηA_t depends on the accumulated load, rather than switching autonomously.	Davis's piecewise-deterministic Markov processes; autonomous on–off queues.
Thm 3.5 p. 85	$A(p) = 2\psi_{2p}(1/2)$ identifies Kolmogorov's constant with the Koenigs linearising function of the logistic map, and that function is hypertranscendental — so it admits no elementary closed form.	Kolmogorov (1938) for $A(p)$ itself; Koenigs (1884); Becker and Bergweiler (1995) for hypertranscendence.
Thms 4.13–4.14 pp. 103–105	The released count W_t carried as a second coordinate: its mean, second moment and variance at *finite* time, as functions of $I(t)$.	Karlin and Tavaré's killing position is the *eventual* geometric, not a time-dependent release coordinate.
Prop 4.17 p. 106	Every load moment is a polynomial $Q_p(I)$ of degree $p+1$ with roots a, b and leading coefficient $(-1)^p p!\,(\lambda/\delta)^p$ — one economy organising the whole moment hierarchy.	The moments themselves sit inside the 1982 generating function; the organisation is what is added.
Thm 5.10 p. 130	Burst size equals the quasi-stationary law, proved by size-biasing: the weight k makes the burst-time integrand a perfect differential, which telescopes to the endpoint. Remark 5.11 gives a *sufficient* criterion for it; necessity is left open.	Karlin and Tavaré give the killing-position law and the quasi-stationary law separately, in different lemmas.
Props 5.12–5.13 pp. 134–135	Burst timing: at $\mu = 0$ the time to a burst of size n is harmonic in n with a Gumbel limit; with death, a closed mean conditional on bursting.	Classical Yule maxima, not previously written as a law for a lytic single-cell assay.
§ 5.9–5.10 pp. 137, 140	Multi-founder kernels g_k with sub-extensive yield $k + \lambda/\delta$; a chained immediate-transfer process with negative-binomial rupture sizes and interval Laplace transforms.	The branching property is classical; the yield identity, and the failure of the naive free sum, are the additions.
Def 6.1, Eq. 6.11 pp. 157–159	An age-structured within-host system driven by *closed-form* birth–death–catastrophe kernels, with no free production kernel left to fit.	Nelson, Gilchrist, Coombs, Hyman and Perelson (2004), who leave the age-dependent kernel $P(a)$ free.
Thm 6.3 p. 161	A projection of the exponential phase onto an effective pair $(p_{\text{eff}}, d_{I,\text{eff}})$, with closed endpoints — so a fitted burst rate is a property of the phase, not of the cell.	Euler–Lotka renewal theory; Wallinga and Lipsitch on generation intervals.
§ 6.5.6 p. 165	A concrete three-to-two identifiability obstruction: (λ, μ, δ) is not recoverable from population data alone.	Miao, Xia, Perelson and Wu (2011) on structural non-identifiability.

Where	The increment	Ancestor
Prop 7.2 p. 183	A Feynman–Kac representation for the joint two-type occupation hazard, with a mesh-limit argument for the pathwise exponential weight.	Standard first-step decomposition for one type; the joint two-type object is the increment.
Prop 7.9 p. 202	At $\lambda_2 = 0$ the solution is built on *ordinary* Bessel functions J_ρ, Y_ρ, a different boundary of the parameter space.	Antal and Krapivsky's bi-critical case uses *modified* Bessel functions, from a different equation.
Def 8.1, Prop 8.3 pp. 214–216	A replicator–suppressor process in which suppressors are consumed one-to-one and never replenished. The escape wall $i > j$ is then rate-*independent*: a counting argument, no rates enter.	Pilyugin and Antia (2000) limit clearance by *handling time*: an engaged effector is occupied, then returns to the free pool. Their threshold therefore depends on the rates.
Prop 8.7, Cor 8.8 pp. 223–223	Under an integer removal budget ϑ, the surviving fraction is $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$ with threshold $\vartheta^* = \log V_\infty / \log a$.	Geometric stop-loss is actuarial folklore; not found for a catastrophe process under an integer immune budget.
Prop 8.12 p. 225	Under threshold removal the flooding criterion *reverses*: bursting now wins exactly when $L < 1$. The same dimensionless group orders the comparison in both directions.	Yuan and Allen (2011) find the ranking of budding and bursting to depend on whether the immune response is activated, by a different mechanism.
Cor 9.3 p. 237	A two-layer clock: speciation intensity tracks unused *individual* capacity through a separate logistic abundance layer, not the species count. The limit law is geometric, with a variance not previously recorded.	Kendall (1948) time-dependent Yule; diversity-dependent models act on species number.
Eq. 9.21 p. 244	A closed form for the moving unstable critical point of the Pimentel resistance chain: a reversal is that point crossing the leading population.	Pimentel (1965) supplies the chain; no antecedent was identified for the critical point.

3. Inherited, and cited as such

Naming these is what makes the two sections above legible as increments. **Karlin and Tavaré (1982)** supply the killed birth–death process itself: the linear-fractional generating function, the closed forms $I(t)$, $D(t)$, $J(t)$, $K(t)$, the roots a, b, the geometric occupancy and the geometric killing position. **Brockwell, Gani and Resnick (1982)** is the more general neighbour, admitting immigration and random catastrophe sizes. **Kolmogorov (1938)** and **Yaglom (1947)** give the constant $A(p)$, the critical decay $S_n \sim 2/n$ and the conditioned limit law; **Pakes (2026)** is the current word on whether that constant admits an explicit form. **Kendall (1948)** gives the linear birth–death generating function reused throughout, and the time-dependent Yule process of Chapter 9. **Antal and Krapivsky (2011)** give the triangular two-type process and its Gauss reduction. **Perelson (1996)**, **Nowak and Bangham (1996)** and **McLean (1993)** give constant-rate viral replication; **Pearson, Krapivsky and Perelson (2011)** give

the invariance of R_0 under matched lifetime yield and the establishment comparison; **Komarova (2007)** gives flooding as a biological idea, and **Yuan and Allen (2011)** a published ranking switch under an immune term. **Nelson et al. (2004)** give the age-structured system; **Miao et al. (2011)** non-identifiability; **Pilyugin and Antia (2000)** finite nonspecific killers. **Nee, May and Harvey (1994)** give the push of the past, against a large diversity-dependent literature. **Ke, Chen and Yang (2013)** give the temperature- and pH-dependent account of the *Y. pestis* switch.

4. Scope of the claims

"Not found" is not a proof of absence. Where the evidence was thin the claim was weakened rather than strengthened, and every statement in Section 1 of this index has a published ancestor supplying the method — so each is written as an original identity obtained by specialising a known technique. Three things are deliberately *not* claimed: the killed birth–death process and its closed forms, which are Karlin and Tavaré's; the constant $A(p)$, which is Kolmogorov's; and the comparison of bursting with budding as a question, which is Pearson's and Komarova's. What is claimed is the calculus built on the first, the Koenigs dictionary for the second, and the criterion L that settles the third in both directions.

Statement on the use of generative AI

I have made extensive use of generative AI and agentic coding tools in producing this thesis, and it is better to be forthcoming and candid about that than to leave it implicit. We live in a rapidly changing digital landscape that is increasingly challenging preconceptions, and it falls to our generation to help define the standard for how AI is taught and integrated into higher education. To that end, I give a summary below of how I used AI in the writing of this thesis, with a chapter-by-chapter account following it.

Where AI produced mathematics, prose or code, I have checked it and I take full responsibility for the content of this thesis.

Account of generative AI use

The following is a short description of how generative AI was used to assist the production of each chapter.

Chapter 1. There was a full introduction in the pre-AI draft of 2024. After the other nine chapters had all been fully redrafted, Claude was used quite simply to modify the existing introduction to reflect the changes in presentation of the new chapters.

Chapter 2. Chapter 2 was partially taken from the pre-AI draft, and partially added to with agent tools. All of the MOC content and much of the branching process definition was in the original. But the Markov Chains section was largely generated.

Chapter 3. The Koenigs result was initially determined by ChatGPT in January 2026. At that time, LLMs were not quite good enough to write sufficient quality prose for the thesis, so I wrote up the whole content of this chapter myself. Later, in July 2026, AI was more capable and I used Claude to rework the existing content into a better flow, and also used Codex to build the fractal generation web tool.

Chapter 4. The original chapter was re-worked in a style pass. The same content from the pre-AI draft was presented more coherently with a small addition of the D_t variable.

Chapter 5. The old BDC extension chapter was divided into what are now Chapters 5 and 6. Chapter 5 is largely the same story as Chapter 4, in that the same content was re-structured and presented better.

Chapter 6. This chapter extends the old "BDC extension" chapter, whose content was clearly unresolved. Among the key additions are the BMVR-style bursting equations, and the L parameter flooding analysis. Practically all of this additional content has been generated autonomously with LLMs. And for this chapter more than any other, the same can be said of the LaTeX write-up.

Chapter 7. The crux of this chapter was present in the pre-AI draft. I had previously shown that the two-type BDC system can be solved with the specified method. I had merely neglected to actually write down the final solution explicitly. There was some complication to that task, but Claude did it effortlessly, and restructured my prose along the way.

Chapter 8. The existing content was restructured and styled with LLM passes.

Chapter 9. Content was trimmed and rearranged into a variable time model collection.

Chapter 10. The quadrant argument was written by me, apart from its figure; everything else in the chapter was generated automatically.

I have used Claude Code (Pro), Grok Build (SuperGrok), Codex (Plus), and Zcode (GLM Coding Lite). Claude and Codex are the most capable models; Grok and Zcode are cheaper and faster, but still reliable alternatives for more mechanical tasks. The originality index was fully researched, verified, and written up by Grok and Claude.

The multiplicative dissipation cellular automaton (CA) in Chapter 9 was agentically coded end-to-end. But other substantial programs were written by me. Among these is the agent-based simulation shown in Chapter 1 (made with p5.js).

There is also the burst suppression simulator and the Pimentel CA. Both of these were fully implemented in C++ using SFML (Simple and Fast Multimedia Library), then later ported to JavaScript and given a new user interface.

Overall, I can say that AI has been remarkably helpful in producing a final result greatly superior to the pre-AI draft. And that generally, LLMs and coding agents have been really enjoyable to use and have recently become surprisingly capable.

If you would like to know more about how I have been using AI, please feel free to get in touch, and I would be happy to talk about it with you.